

# Mathematical Reality and Physical Actuality

## *Resolving the Ontological Controversy in Quantum Foundations*

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### Abstract

At the 2025 Helgoland centenary conference, Anton Zeilinger and Alain Aspect exchanged a philosophically consequential disagreement. Zeilinger asserted that there is no quantum world and that the quantum state is nothing more than information; Aspect replied that quantum states are real and objective. A subsequent Nature survey of over 1,100 physicists confirmed that this disagreement is structural and persistent. A previous study (Zhong, 2026a) demonstrated that this divergence cannot be resolved within a single-layer ontological framework. The present paper completes that argument by identifying the precise conceptual distinction the earlier analysis left open: the wavefunction is mathematically real and objectively defined, but it describes the probability structure of possible physical outcomes rather than a simultaneously existing classical reality. Measurement outcomes belong to the domain of physically actualised facts. The Zeilinger position captures an essential feature of the wavefunction domain; the Aspect position captures an essential feature of the outcome domain. Their apparent contradiction arises from the conflation of mathematical reality with physical actuality—a conflation all five major interpretations of quantum mechanics share. The formal interface between these two levels is expressed by the staged mapping  $|\psi\rangle \rightarrow C_i \rightarrow T_{\mu\nu}(C_i) \rightarrow G_{\mu\nu}$  (Zhong, 2026b), which specifies the conceptual relationship between quantum probability structures and classical physical descriptions. This two-level definition dissolves the ontological controversy in quantum foundations.

**Keywords:** quantum mechanics interpretation; wavefunction ontology; mathematical reality; physical actuality; measurement problem; Zeilinger–Aspect divergence; ontological stratification; quantum gravity interface

## **1. Introduction: The Dispute and Its Stakes**

In June 2025, the physics community gathered on the German island of Helgoland to mark one hundred years of quantum mechanics. The exchange that crystallised the occasion was brief. Anton Zeilinger, co-recipient of the 2022 Nobel Prize in Physics, stated: "There is no quantum world." Alain Aspect, joint recipient of the same prize for his landmark Bell inequality tests, replied: "I disagree." A Nature survey of more than 1,100 physicists, published on 30 July 2025, subsequently confirmed that this disagreement is not a personal philosophical difference but a structural and unresolved division within the research community (Gibney, 2025). One hundred years after the birth of quantum mechanics, physicists agree on the mathematical formalism and disagree, profoundly, on what that formalism describes.

A previous study (Zhong, 2026a) argued that this disagreement cannot be resolved within a single-layer ontological framework—that is, within any framework that assumes the wavefunction and measurement outcomes belong to the same type of being and must be accommodated within a single set of ontological categories. That paper further argued that a minimal ontological stratification, distinguishing the domain of quantum states from the domain of measurement outcomes, is a necessary condition for rendering the Zeilinger and Aspect positions simultaneously coherent. It did not, however, provide a precise characterisation of what distinguishes the two domains or a formal account of the interface between them.

The present paper completes that argument. Its central claim is that the Zeilinger–Aspect divergence arises from a specific conceptual conflation: the conflation of mathematical reality with physical actuality. The wavefunction is mathematically real and objectively defined—it is not a subjective probability assignment, not a mere calculational device, and not an expression of observer ignorance. But it describes the probability structure of possible physical outcomes for a single system, not a collection of simultaneously existing classical realities. Measurement outcomes, by contrast, are physically actualised facts: definite, observer-independent, and characterised by the non-local correlations that Aspect's experiments have established. Zeilinger is correct about the wavefunction domain; Aspect is correct about the outcome domain. The two positions are not contradictions. They are accurate descriptions of

two different aspects of the same physical situation, which have been forced into apparent conflict by the assumption that they must occupy the same ontological level.

The paper proceeds as follows. Section 2 reconstructs the Zeilinger and Aspect positions with precision, identifying exactly what each physicist is responding to in the quantum mechanical formalism. Section 3 argues that all five major interpretations documented in the Nature survey share a common conceptual conflation: the conflation of mathematical reality with physical actuality, in one direction or the other. Section 4 develops the distinction between mathematical reality and physical actuality as the conceptual key to dissolving the divergence. Section 5 introduces the staged mapping  $|\psi\rangle \rightarrow C_i \rightarrow T_{\mu\nu}(C_i) \rightarrow G_{\mu\nu}$  as the formal expression of the interface between the two levels, drawing on the companion paper on quantum gravity (Zhong, 2026b). Section 6 addresses objections and clarifications. Section 7 concludes.

## **2. What Each Physicist Is Responding To**

### **2.1 Zeilinger and the Informational Character of the Wavefunction**

Zeilinger's position is a response to genuine and well-established features of the wavefunction as it appears in the quantum mechanical formalism. The wavefunction  $\psi$  is defined on the configuration space of a physical system—an abstract, high-dimensional mathematical space, not the three-dimensional space of ordinary physical experience. It assigns probability amplitudes to possible measurement outcomes, not definite values to physical properties. When a measurement is performed and an outcome is obtained, the wavefunction is updated: what was a superposition of possible outcomes becomes, in effect, a representation of the one outcome that was realised. The wavefunction, in Zeilinger's reading, is therefore essentially informational: it encodes what can be predicted about measurement results, not what is the case independently of any measurement (Zeilinger, 1999; Fuchs, Mermin and Schack, 2014).

This reading is grounded in the mathematical structure of the theory. The wavefunction does not assign definite values to position and momentum simultaneously; it assigns probability distributions. It does not describe a particle at a location; it describes the probability that a particle will be found at a location if a measurement is performed. These features are not artefacts of interpretation; they are features of the mathematical object. The informational reading is a natural and defensible response to them.

Where Zeilinger's position becomes philosophically contentious is in its further inference: that because the wavefunction is informational in character, there is no quantum world—no mind-

independent domain of physical reality to which the wavefunction corresponds. This inference moves from the correct observation that the wavefunction does not describe definite classical properties to the stronger conclusion that there is nothing for the wavefunction to be about. It is this inference, as the present paper argues, that is mistaken.

## **2.2 Aspect and the Objectivity of Quantum Correlations**

Aspect's position is an equally well-grounded response to a different set of features of quantum mechanics. His experimental work in the 1980s, building on Bell's 1964 theoretical analysis, produced the first convincing demonstration that quantum correlations violate Bell inequalities (Aspect, Grangier and Roger, 1982; Bell, 1964). This result establishes that the statistical predictions of quantum mechanics cannot be reproduced by any theory that assumes both locality and realism in the classical sense: that particles have definite properties prior to measurement, and that those properties are determined by factors confined to their immediate vicinity.

For Aspect, this experimental legacy grounds a realist commitment. The correlations between entangled particles are objective, reproducible, and non-local. They cannot be explained as the product of pre-existing local properties—Bell's theorem rules that out. They cannot be explained as the joint updating of subjective beliefs—the correlations obtain whether or not any agent is present or aware of them. Something in the world must account for these correlations, and that something must be non-local in character. The wavefunction, or whatever the wavefunction describes, must have ontological weight: it must correspond to something real, not merely to an observer's bookkeeping.

This position is equally well-supported. The Bell experiments are among the most precisely confirmed results in the history of physics, and the non-local correlations they establish are a structural feature of quantum mechanics that no purely epistemic interpretation can fully account for without invoking some form of objective, non-local physical structure. Aspect is correct that the outcomes of quantum measurements, and the correlations between them, are objectively real.

## **2.3 The Structural Asymmetry**

What is philosophically significant about the exchange is that both physicists are responding accurately to genuine features of the same theory, and those features do not belong to the same level of description. Zeilinger is responding to the character of the wavefunction as a

mathematical object: its probabilistic, contextual, and high-dimensional character. Aspect is responding to the character of measurement outcomes and their correlations: their definiteness, reproducibility, and objective non-locality. The conflict arises not because one physicist is wrong but because both are correct about different aspects of quantum mechanics, and the single-layer ontological assumption forces those Aspect's onto a single canvas where they appear to contradict each other.

The question the present paper pursues is whether this conflict is genuine—whether one position must be abandoned—or whether it is an artefact of a shared conceptual conflation. The answer, it will be argued, is the latter.

### **3. The Common Conflation of the Five Major Interpretations**

The 2025 Nature survey documented the distribution of interpretive preferences across five major positions: the Copenhagen interpretation (favoured by 36% of respondents), many-worlds (15%), Bohmian mechanics (7%), spontaneous collapse theories (4%), and epistemic or QBist approaches (17%), with a substantial minority selecting other formulations. Each of these interpretations represents a sophisticated and serious attempt to account for the full range of quantum mechanical phenomena within a coherent conceptual framework. None, this section argues, succeeds—and the pattern of failure is the same in each case: the conflation of mathematical reality with physical actuality, in one direction or the other.

The Copenhagen interpretation, in its standard formulation, holds that the wavefunction represents the probabilities of measurement outcomes and that asking about the state of a system between measurements is unanswerable in principle. It combines elements of both the Zeilinger and Aspect positions—taking the wavefunction as non-referential while treating measurement outcomes as real—without providing a principled account of the relationship between them. The result, as Bohr's defenders acknowledge, is a systematic refusal to answer the question of what the wavefunction is about (Bohr, 1935). That approximately 29% of self-described Copenhagen adherents in the survey simultaneously believe the wavefunction represents something real illustrates the internal inconsistency: the interpretation is used as a practical tool without its philosophical content being clearly apprehended (Gibney, 2025).

The many-worlds interpretation takes the wavefunction with maximum ontological seriousness: it treats the wavefunction as the totality of physical reality, evolving unitarily at all times, with all branches equally real (Everett, 1957). This accommodates the Aspect intuition that the

wavefunction is ontically real but at the cost of generating a severe problem at the other end: if all branches are real, the appearance of a single definite outcome in any given branch requires explanation within a framework that treats all outcomes as simultaneously existing. The classical world of definite facts becomes a secondary, emergent phenomenon within a fundamentally quantum ontology. Many-worlds mistakes the full mathematical structure of the wavefunction for the full ontological structure of physical reality.

Bohmian mechanics assigns definite trajectories to particles, guided by a real pilot wave (Bohm, 1952). It is explicitly psi-ontic and accommodates Aspect's realism. But it achieves this by introducing a wavefunction defined in the configuration space of all particles—not in ordinary three-dimensional physical space—and by treating this high-dimensional mathematical object as a real physical field. The ontological status of a field defined in a space that is not the space of physical experience remains philosophically opaque. Bohmian mechanics mistakes a mathematical structure for a physical entity without clarifying what kind of entity could exist in configuration space.

Spontaneous collapse theories modify the Schrödinger equation to introduce a physical collapse mechanism (Ghirardi, Rimini and Weber, 1986). They are empirically testable in principle and represent the most honest attempt to provide a physical account of the domain transition between quantum superposition and definite classical outcomes. But they add a stochastic collapse term that is physically unexplained, introduced specifically to solve the measurement problem. The collapse mechanism is an ad hoc mathematical modification that gestures at a physical process without identifying one.

Epistemic and QBist interpretations hold that the wavefunction encodes an agent's degrees of belief about measurement outcomes, not a description of physical reality (Fuchs, Mermin and Schack, 2014; Caves, Fuchs and Schack, 2002). They handle the Zeilinger dimension with elegance but face a severe difficulty with the Aspect dimension: if the wavefunction is purely subjective, what accounts for the objective, non-local correlations between distant entangled particles? The correlations cannot be attributed to pre-existing local properties—Bell's theorem rules that out—and cannot be fully explained as the joint updating of different agents' beliefs without some shared objective physical structure. Epistemic interpretations mistake mathematical instrumentalism for ontological adequacy.

The pattern of failures is not random. Each interpretation fails specifically at the point where it must account for phenomena from one domain using the conceptual resources of another. The

failure is always at the same structural location: the interface between the domain of quantum probability structures and the domain of physically actualised facts. This repeated failure at the same point is precisely what would be expected if the conflation of mathematical reality with physical actuality is the source of the difficulty.

## **4. Mathematical Reality and Physical Actuality: The Conceptual Key**

### **4.1 The Distinction**

The central conceptual distinction required to dissolve the Zeilinger–Aspect divergence is the following: mathematical reality and physical actuality are not the same thing, and quantum mechanics describes both, at different levels.

Mathematical reality, as used here, refers to the objective, mind-independent character of the wavefunction as a mathematical object. The wavefunction is not a subjective probability assignment, not an expression of observer ignorance, and not a mere calculational device. It is a real description of a real feature of a physical system: the complete probability structure of possible outcomes for that system under specified conditions. This probability structure is not created by the observer; it is a property of the system, determined by its physical constitution and its history of interactions. It is objective in the same sense that any mathematical description of a physical situation is objective: it is the same for all observers, it evolves according to deterministic laws (the Schrödinger equation), and it generates precise quantitative predictions that can be confirmed or disconfirmed by experiment.

Physical actuality, by contrast, refers to the domain of definite, realised physical facts: the definite position of a particle after measurement, the definite correlation between two entangled particles after their joint measurement, the definite reading of a detector. These are facts in the full classical sense: they are either the case or not the case, they are observer-independent, and they are the objects to which general relativistic descriptions of spacetime—the Einstein field equations—apply. Physical actuality is the domain of what has happened, not of what could happen.

The wavefunction belongs to the domain of mathematical reality. It describes what could happen—the full probability structure of possible outcomes—for a system that has not yet undergone the relevant physical interaction. Measurement outcomes belong to the domain of physical actuality. They describe what has happened—the definite result of a physical interaction that has already occurred.

These two domains are not competing descriptions of the same moment. They are descriptions of different aspects of the same physical situation: the wavefunction describes the situation before a specific outcome is actualised; the measurement outcome describes the situation after. The transition between them is not a process within the wavefunction formalism—the Schrödinger equation does not generate it—but the actualisation of one possibility from among those encoded in the probability structure.

#### **4.2 Why This Is Not the Epistemic Interpretation**

It is essential to distinguish the present account from the epistemic or QBist interpretation, with which it shares some surface features. Both accounts hold that the wavefunction does not describe a collection of simultaneously existing classical realities. But they differ fundamentally on the question of whether the wavefunction is objective.

On the epistemic interpretation, the wavefunction represents an agent's subjective degrees of belief. It is updated by measurement because the agent has acquired new information, not because anything physical has changed. The wavefunction is, in the strict sense, about the agent, not about the world.

On the present account, the wavefunction is objectively real: it is a real feature of a physical system, not of any observer's mental state. Its probabilistic character is not a sign of epistemic limitation but of the genuine physical character of the pre-actualisation domain. The wavefunction is updated by physical interaction not because an agent has acquired information but because a physical process has occurred that actualises one of the possible outcomes encoded in the probability structure. The update is ontological, not epistemic: something in the world has changed, not merely in the agent's knowledge.

This distinction matters for the Aspect problem. If the wavefunction is objective—if it describes a real feature of physical systems—then the non-local correlations between entangled particles can be grounded in the objective probability structure that both particles share. The correlations are objective because the wavefunction is a mathematically calculated objective probability structure; they are non-local because the wavefunction, as a description of the probability structure of a composite system, is defined over the composite system's configuration space, which is not decomposable into independent local components. Aspect is satisfied: the correlations are grounded in something real, objective, and physical.

At the same time, Zeilinger is satisfied: the wavefunction does not describe a classical world of definite simultaneous properties. It describes the probability structure of possible outcomes for a system that has not yet actualised one of them. "There is no quantum world" in the sense of a collection of simultaneously existing classical facts—correct. But there is a quantum domain: a domain of objective probability structures, mathematically real, physically significant, and causally efficacious in determining the statistics of measurement outcomes.

### **4.3 What the Distinction Resolves**

With this distinction in place, the Zeilinger–Aspect divergence dissolves without remainder.

Zeilinger is correct that there is no quantum world in the sense of a classical reality with definite properties existing prior to and independently of measurement. The wavefunction domain is not a classical domain. It is a domain of objective probability structures: real, but not yet actualised into definite classical facts. The informational character of the wavefunction that Zeilinger emphasises is genuine: the wavefunction does encode information about possible outcomes, and it is correctly described as a probability amplitude structure rather than a catalogue of simultaneously existing classical states.

Aspect is correct that the wavefunction describes something real and that the correlations between entangled particles are objective, non-local, and not derivable from any account that treats the wavefunction as merely subjective. The wavefunction domain is real, objective, and causally efficacious. The non-local correlations are grounded in the objective probability structure of the composite quantum system, which is not decomposable into independent local components. Realism about the wavefunction is appropriate, provided it is understood as realism about a domain of objective probability structures rather than realism about a domain of simultaneously existing classical facts.

The two positions are not contradictions. They are accurate descriptions of two different domains—the domain of objective probability structures and the domain of physically actualised facts—which the single-layer ontological assumption has forced into apparent conflict by treating both as descriptions of the same kind of entity.

## **5. The Formal Interface: The Staged Mapping**

The conceptual distinction between mathematical reality and physical actuality has a formal expression. In the companion paper on quantum gravity (Zhong, 2026b), the interface between

the quantum description of matter and the classical description of spacetime is expressed as a staged mapping:

$$|\psi\rangle \rightarrow C_i \rightarrow T_{\mu\nu}(C_i) \rightarrow G_{\mu\nu}$$

Here  $|\psi\rangle$  is the quantum state of a physical system, encoding its complete probability structure over possible configurations.  $C_i$  is a classically realisable configuration—an actualised outcome, a definite physical fact.  $T_{\mu\nu}(C_i)$  is the energy-momentum tensor corresponding to that configuration, and  $G_{\mu\nu}$  is the spacetime geometry determined by that tensor through the Einstein field equations.

The mapping is staged rather than direct. The standard framing of the quantum gravity problem assumes a direct mapping  $|\psi\rangle \rightarrow g_{\mu\nu}$ : the quantum state is supposed to immediately and directly determine a spacetime geometry. This assumption generates the apparent inconsistency, because a superposed quantum state cannot determine a unique classical metric. The staged mapping resolves the inconsistency by recognising that the spacetime geometry does not correspond to the quantum probability structure  $|\psi\rangle$  but to the actualised configuration  $C_i$  that the probability structure is a description of. The quantum description and the gravitational description apply at different levels of the physical situation. The mapping is not proposed as a dynamical equation but as an ontological decomposition of the transition between levels.

This formal structure directly encodes the conceptual distinction of the preceding section. The transition from  $|\psi\rangle$  to  $C_i$  is the transition from mathematical reality to physical actuality: from the objective probability structure of possible outcomes to the definite physical fact that is realised through physical interaction. The transition from  $C_i$  to  $T_{\mu\nu}(C_i)$  to  $G_{\mu\nu}$  is the application of the gravitational description to the actualised configuration—the domain in which classical and relativistic physics applies.

For the purposes of the Zeilinger–Aspect dispute, the relevant portion of the staged mapping is the first transition:  $|\psi\rangle \rightarrow C_i$ . This transition is the domain transition from the wavefunction domain to the outcome domain. Zeilinger’s position correctly characterises the  $|\psi\rangle$  side of this transition: the quantum state is a probability structure, not a classical world. Aspect’s position correctly characterises the  $C_i$  side: the actualised outcome is real, definite, and objectively correlated with the outcomes of measurements on entangled systems. The transition itself—the actualisation of a definite outcome from a quantum probability structure—is what the present

framework identifies as the physically significant event, and what all five major interpretations have been attempting, in their different ways, to account for.

The staged mapping does not specify what physical process constitutes the transition from  $|\psi\rangle$  to  $C_i$ . This remains the central open question of quantum foundations. What the mapping does specify is the conceptual structure that any account of that process must respect: the quantum state and the actualised outcome are not entities of the same type, and the transition between them is not a process within the Schrödinger equation's domain but the boundary of that domain. Candidate physical accounts of the transition include objective collapse models (Ghirardi, Rimini and Weber, 1986), gravitational collapse proposals (Penrose, 1996), and decoherence-based accounts (Zurek, 2003; Schlosshauer, 2005). Each of these is a proposal about the physical mechanism of the  $|\psi\rangle \rightarrow C_i$  transition; none is in conflict with the conceptual framework proposed here.

## **6. Objections and Clarifications**

### **6.1 Is This Just Copenhagen?**

The present account shares with the Copenhagen interpretation the claim that the wavefunction does not describe a classical world of definite simultaneously existing properties. But it differs from Copenhagen in two important respects. First, it offers a positive characterisation of what the wavefunction does describe: an objective probability structure over possible physical outcomes. Copenhagen declines to answer this question. Second, it does not require a classical/quantum cut imposed from outside the formalism, nor does it make the measurement outcome observer-dependent. The actualisation of an outcome is a physical process, not an epistemic event. The present framework is realist about both domains: the wavefunction domain and the outcome domain are both objectively real, each at its own level of description.

### **6.2 Does This Solve the Measurement Problem?**

It does not solve the measurement problem in the sense of providing a dynamical mechanism for the  $|\psi\rangle \rightarrow C_i$  transition. It dissolves the measurement problem as classically formulated—as the problem of how the Schrödinger equation generates a definite classical outcome—by showing that this formulation is malformed: the Schrödinger equation does not generate outcomes because outcomes are not within its domain. The correctly formulated question is what physical process constitutes the domain transition, and this question remains open. The present framework identifies what kind of answer would be adequate: a physical account of the

transition from the domain of objective probability structures to the domain of physically actualised facts.

### **6.3 What About the Preferred Basis Problem?**

The preferred basis problem—the question of why the universe branches or decoheres in one basis rather than another—arises most acutely within the many-worlds interpretation. Within the present framework, the question takes a different form: in what basis does the domain transition from  $|\psi\rangle$  to  $C_i$  occur? This is a genuine physical question, and the present framework does not answer it. Decoherence theory identifies the conditions under which certain bases become dynamically preferred through environmental interaction (Zurek, 1981; Joos et al., 2003), and these conditions are relevant to identifying the physical regime in which domain transitions occur. The present framework is compatible with decoherence as an account of the approach to the domain boundary; it does not specify the mechanism of the crossing.

### **6.4 Is Mathematical Reality a New Ontological Category?**

The distinction between mathematical reality and physical actuality is not a new ontological category introduced ad hoc for this problem. It is a distinction that is implicit in the structure of quantum mechanics and that has been partially recognised by several existing accounts. Heisenberg's notion of *potentia*—the wavefunction as describing objective tendencies or potentialities that are real features of the world but not actualities until realised through interaction—is the closest historical anticipation (Heisenberg, 1958). The present account makes this distinction explicit, characterises it precisely, and shows that it is sufficient to dissolve the Zeilinger–Aspect divergence without introducing any additional physical structure.

## **7. Conclusion**

One hundred years after the birth of quantum mechanics, the Zeilinger–Aspect divergence at the Helgoland centenary conference is not a sign of conceptual stagnation. It is a precise diagnostic: it marks, with unusual clarity, the internal boundary of the current theoretical framework. Both physicists are correct. Each is describing a genuine feature of quantum mechanics. The conflict between their descriptions arises from a shared conceptual conflation—the conflation of mathematical reality with physical actuality—that is embedded in all five major interpretations of quantum mechanics documented by the 2025 Nature survey.

The wavefunction is mathematically real and objectively defined. It describes the complete probability structure of possible outcomes for a physical system. This probability structure is a real feature of the world, not a representation of observer ignorance or a subjective degree of belief. Zeilinger is correct that this probability structure is not a classical world of simultaneously existing definite properties. Aspect is correct that it is objectively real and that the non-local correlations it generates are objective, physical facts. The two positions are accurate descriptions of two different aspects of the same physical situation—the domain of objective probability structures and the domain of physically actualised facts—which the single-layer ontological assumption has forced into apparent conflict.

The formal interface between these two domains is expressed by the staged mapping  $|\psi\rangle \rightarrow C_i \rightarrow T_{\mu\nu}(C_i) \rightarrow G_{\mu\nu}$ , developed in the companion paper on quantum gravity (Zhong, 2026b). This mapping shows that the quantum description and the classical description of spacetime apply at different levels of the physical situation and that their apparent inconsistency dissolves once the staged character of the interface is recognised. The same conceptual move that resolves the quantum gravity problem at the level of general relativity resolves the Zeilinger–Aspect dispute at the level of quantum foundations: recognising that mathematical reality and physical actuality are descriptions that apply to different aspects of the physical situation, connected by a domain transition that neither the Schrödinger equation nor the Einstein field equations can generate from within their own domains.

Robert Spekkens observed in the Nature survey that "once we find the correct interpretation, it will announce itself by virtue of offering more coherence than anything before" (Gibney, 2025). The present paper does not claim to have found the correct interpretation in the sense of providing a dynamical mechanism for the domain transition. It claims to have identified the conceptual distinction—between mathematical reality and physical actuality—that is the minimum requirement for any coherent interpretation, and to have shown that this distinction dissolves the most persistent and public disagreement in the foundations of quantum mechanics. The Zeilinger–Aspect divergence, viewed in this light, is not embarrassing. It is informative. It tells us, with precision, what the next theoretical advance must accomplish: not a new equation, but a clear account of the physical process by which the domain of objective probability structures gives rise to the domain of physically actualised facts.

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